

PATRICK S. KEOUGH

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AI evaluation researcher and builder. I find out what language models actually do at scale and ship the evidence: ten studies, 140,000+ scored observations, two arXiv preprints, each released with its pipeline code and a receipt for every claim. I also build the systems: an open-source voice-cloning MCP server (11,000 lines, 153 tests), a 23,000-line retrieval memory system, and contracted software for two Sydney clients.

Experience

Independent AI Evaluation Researcher, Sydney

Jun 2025 to present

- Designed and ran ten studies of language models totalling 140,000+ scored observations; the largest ran 28,800 simulated patients across four model families and 120 demographic cohorts against survey-weighted NHANES anchors.
- Built every LLM-as-judge pipeline end to end: multi-axis severity rubrics, cross-model judge validation, position and self-preference bias controls, at scale over OpenRouter and native provider APIs.
- Validated automated scoring against 236 human annotations (Cohen's $\kappa = 0.78$ between the judge and the human consensus; 0.58 between human raters).
- Statistics in Python: mixed-effects models, non-parametric tests, effect sizes, inter-rater reliability, BH-FDR and Holm correction. Two preprints on arXiv, one co-authored paper with ACU in preparation.

AI Systems Contractor, Advanced Human Technologies (AHT Group), Sydney

Jun 2026 to present

- Built and delivered a voice-emulation pipeline for a client corpus: LLM-as-judge rubric, a measured quality gate every draft must clear, shipped as an MCP server with hybrid retrieval. Contracted and invoiced.

Technical Consultant, Leadership diagnostics for a private client, Sydney

2026

- Built a company-evaluation web app (5,500 lines of Python, Vite front end) that scrapes public data, scores it against three published leadership frameworks with a source-verification pass, and renders an editable executive briefing with PDF export.
- Built a Gmail classification and routing system in Apps Script for the same client: 77 unit tests and 37 integration tests against an in-memory Gmail, with structural guarantees that it cannot send, filter, or delete.

Teaching Assistant, Statistics in Psychology (PSYCH 315), University of Washington

Winter 2025

Systems

Mimesis (open source, github.com/pskeough/Mimesis_Voice_Clone). Stylometric voice-cloning MCP server, 11,231 lines of Python, 153 tests. Selects drafts on a 13-feature author fingerprint instead of an LLM judge; its cadence feature set separates real prose from rhythm-destroyed prose at AUROC 1.000 where an order-invariant set performs at chance.

Claude Mind. Local memory and retrieval system for an agentic coding tool, 23,076 lines of TypeScript: SQLite with sqlite-vec, bi-encoder recall, cross-encoder rerank, Personalized PageRank graph channel, 17 MCP tools, 2,475 captured sessions. Audited end to end: 0 scope leaks on 58 probes, p50 retrieval 3.7 s.

Signal. Nightly multi-agent pipeline that fans out five collector agents, dedupes against a persistent index, ranks against a feedback-tuned profile, and publishes an offline PWA; 31 consecutive nightly runs.

Selected research

The Granularity Gap. arXiv:2606.05183, v3. Severity audit of 8,830 responses across three Gemini generations, judge validated on 236 human annotations. A binary refuse-or-comply verdict explains 29% of the variance in graded severity, so pass/fail and graded metrics measure different things.

Plausible Patients, Impossible Populations. arXiv:2604.17359, v2. 28,800 simulated assessments, four models, 120 cohorts. Every benchmarkable group is inflated against NHANES while 97.3% of single cases pass a DSM-5 gateway rule, so case-level review cannot detect the population-level failure.

Eight further studies, including **A Validity Ladder for LLM-Simulated Clinical Populations** (in preparation for *Behavior Research Methods*). Code and data at github.com/pskeough.

Skills

Evaluation and statistics. LLM-as-judge design, severity rubrics, cross-model validation, bias controls, human validation and IRR, mixed-effects models, effect sizes, multiple-comparison correction, survey-weighted analysis, red-teaming. **ML and LLM**. Large-N inference pipelines, RAG and hybrid retrieval, embeddings and rerankers, LoRA/SFT/DPO fine-tuning and adapter serving, multi-GPU and rented-GPU training, ablation, SAE interpretability, stylometry, local inference (llama.cpp), MCP servers, agentic workflows. **Engineering**. Python (pandas, NumPy, SciPy, statsmodels), TypeScript, React, Flask, SQLite, pytest, Git, LaTeX.

Education and leadership

BA Psychology, University of Washington. Conferred June 2025. GPA 3.77/4.00. Dean's List every quarter.

Students for Sensible Drug Policy. National Executive Policy Board, one of two student representatives (2022 to 2023). Founded the first naloxone distribution station at UW; ran overdose-response trainings with the Washington Department of Health.